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XY Chart

In the context of mermaid-js, the XY chart is a comprehensive charting module that encompasses various types of charts that utilize both x-axis and y-axis for data representation. Presently, it includes two fundamental chart types: the bar chart and the line chart. These charts are designed to visually display and analyze data that involve two numerical variables.

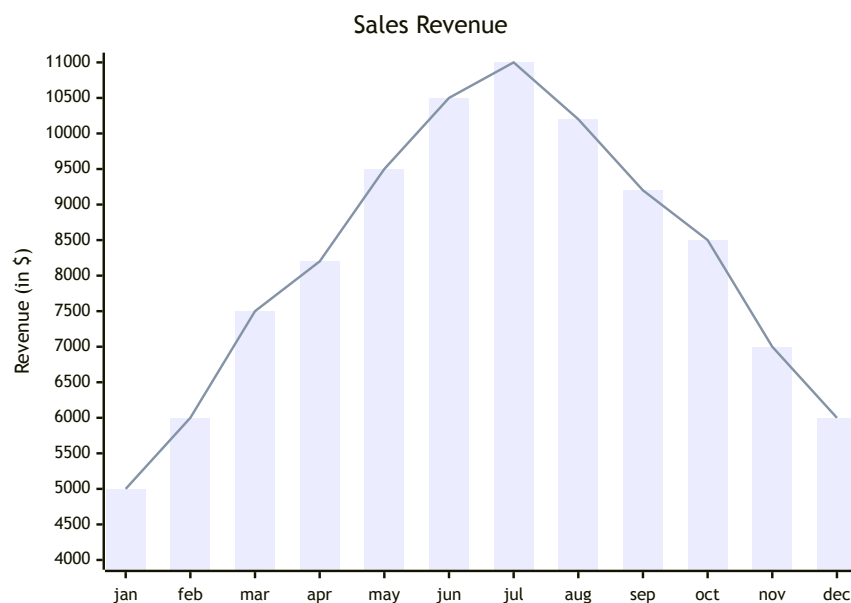
It's important to note that while the current implementation of mermaid-js includes these two chart types, the framework is designed to be dynamic and adaptable. Therefore, it has the capacity for expansion and the inclusion of additional chart types in the future. This means that users can expect an evolving suite of charting options within the XY chart module, catering to various data visualization needs as new chart types are introduced over time.

Example

Code:

```
mermaid
xychart
  title "Sales Revenue"
  x-axis [jan, feb, mar, apr, may, jun, jul, aug, sep, oct, nov, dec]
  y-axis "Revenue (in $)" 4000 --> 11000
  bar [5000, 6000, 7500, 8200, 9500, 10500, 11000, 10200, 9200, 8500, 7000, 6000]
  line [5000, 6000, 7500, 8200, 9500, 10500, 11000, 10200, 9200, 8500, 7000, 6000]
```

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Syntax

INFO

All text values that contain only one word can be written without `"` . If a text value has many words in it, specifically if it contains spaces, enclose the value in `"`

Orientations

The chart can be drawn horizontal or vertical, default value is vertical.

```
xychart horizontal
...

```

Title

The title is a short description of the chart and it will always render on top of the chart.

Example

```
xychart
  title "This is a simple example"
  ...

```

INFO

If the title is a single word one no need to use `"` , but if it has space `"` is needed

x-axis

The x-axis primarily serves as a categorical value, although it can also function as a numeric range value when needed.

Example

1. `x-axis title min --> max` x-axis will function as numeric with the given range
2. `x-axis "title with space" [cat1, "cat2 with space", cat3]` x-axis if categorical, categories are text type

y-axis

The y-axis is employed to represent numerical range values, it cannot have categorical values.

Example

1. `y-axis title min --> max`
2. `y-axis title` it will only add the title, the range will be auto generated from data.

INFO

Both x and y axis are optional if not provided we will try to create the range

Line chart

A line chart offers the capability to graphically depict lines.

Example

1. `line [2.3, 45, .98, -3.4]` it can have all valid numeric values.

Bar chart

A bar chart offers the capability to graphically depict bars.

Example

- 1. `bar [2.3, 45, .98, -3.4]` it can have all valid numeric values.

Simplest example

The only two things required are the chart name (`xychart`) and one data set. So you will be able to draw a chart with a simple config like

```
xychart
  line [+1.3, .6, 2.4, -.34]
```

Chart Configurations

Parameter	Description	Default value
width	Width of the chart	700
height	Height of the chart	500
titlePadding	Top and Bottom padding of the title	10
titleFontSize	Title font size	20
showTitle	Title to be shown or not	true
xAxis	xAxis configuration	AxisConfig
yAxis	yAxis configuration	AxisConfig
chartOrientation	'vertical' or 'horizontal'	'vertical'
plotReservedSpacePercent	Minimum space plots will take inside the chart	50
showDataLabel	Should show the value corresponding to the bar within the bar	false
showDataLabelOutsideBar	If showing data label then show it outside the bar.	false

AxisConfig

Parameter	Description	Default value
showLabel	Show axis labels or tick values	true
labelFontSize	Font size of the label to be drawn	14
labelPadding	Top and Bottom padding of the label	5
showTitle	Axis title to be shown or not	true
titleFontSize	Axis title font size	16
titlePadding	Top and Bottom padding of Axis title	5
showTick	Tick to be shown or not	true
tickLength	How long the tick will be	5
tickWidth	How width the tick will be	2
showAxisLine	Axis line to be shown or not	true
axisLineWidth	Thickness of the axis line	2

Chart Theme Variables

Themes for xychart reside inside the `xychart` attribute, allowing customization through the following syntax:

yaml

```
---
config:
  themeVariables:
    xyChart:
      titleColor: '#ff0000'
---
```

Parameter	Description
backgroundColor	Background color of the whole chart
titleColor	Color of the Title text
dataLabelColor	Color of the Data labels (if shown)
xAxisLabelColor	Color of the x-axis labels
xAxisTitleColor	Color of the x-axis title
xAxisTickColor	Color of the x-axis tick
xAxisLineColor	Color of the x-axis line
yAxisLabelColor	Color of the y-axis labels
yAxisTitleColor	Color of the y-axis title
yAxisTickColor	Color of the y-axis tick
yAxisLineColor	Color of the y-axis line
plotColorPalette	String of colors separated by comma e.g. "#f3456, #43445"

Setting Colors for Lines and Bars

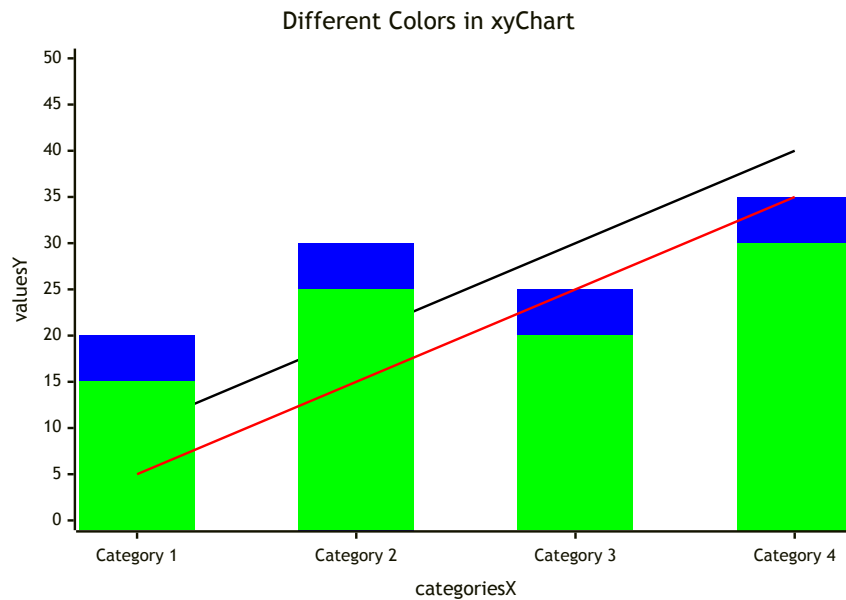
To set the color for lines and bars, use the `plotColorPalette` parameter. Colors in the palette will correspond sequentially to the elements in your chart (e.g., first bar/line will use the first color specified in the palette).

Code:

mermaid

```
---
config:
  themeVariables:
    xyChart:
      plotColorPalette: '#000000, #0000FF, #00FF00, #FF0000'
---
xychart
title "Different Colors in xyChart"
x-axis "categoriesX" ["Category 1", "Category 2", "Category 3", "Category 4"]
y-axis "valuesY" 0 --> 50
%% Black line
line [10,20,30,40]
%% Blue bar
bar [20,30,25,35]
%% Green bar
bar [15,25,20,30]
%% Red line
line [5,15,25,35]
```

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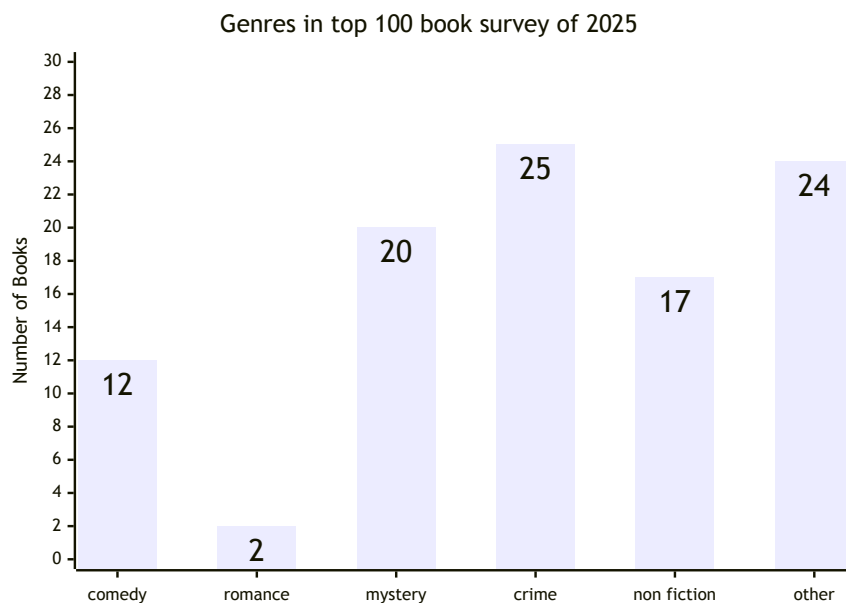


Displaying individual values on a bar chart (v11.14.0+)

To show the value corresponding to a bar specify `showDataLabel: true`.

Code:

```
---mermaid
config:
  xyChart:
    showDataLabel: true
---
xychart
  title "Genres in top 100 book survey of 2025"
  x-axis [comedy, romance, mystery, crime, "non fiction", other]
  y-axis "Number of Books" 0 --> 30
  bar [12,2,20,25,17,24]
```



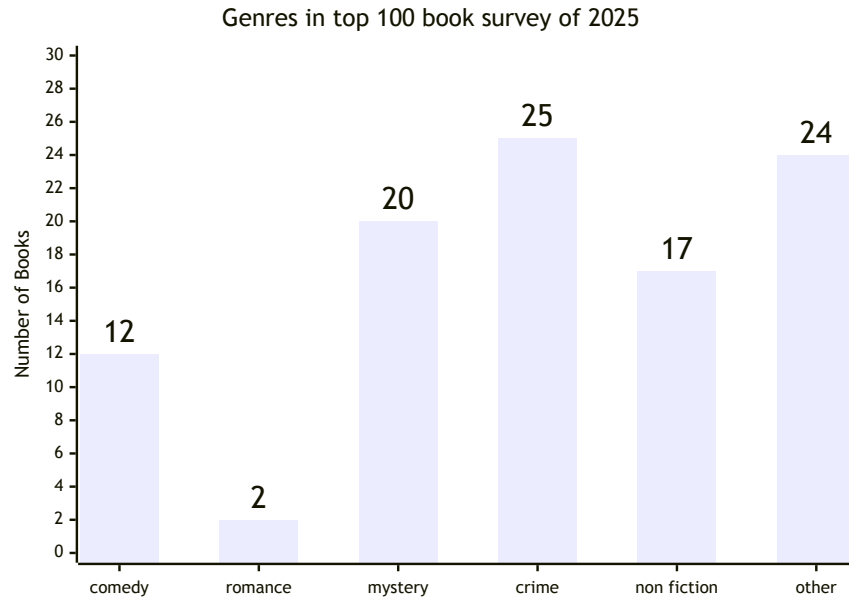
Labels are shown within the bar by default. To show the labels outside the bar, specify `showDataLabelOutsideBar: true`.

Code:

mermaid

```
---
config:
  xyChart:
    showDataLabel: true
    showDataLabelOutsideBar: true
---
xychart
  title "Genres in top 100 book survey of 2025"
  x-axis [comedy, romance, mystery, crime, "non fiction", other]
  y-axis "Number of Books" 0 --> 30
  bar [12,2,20,25,17,24]
```

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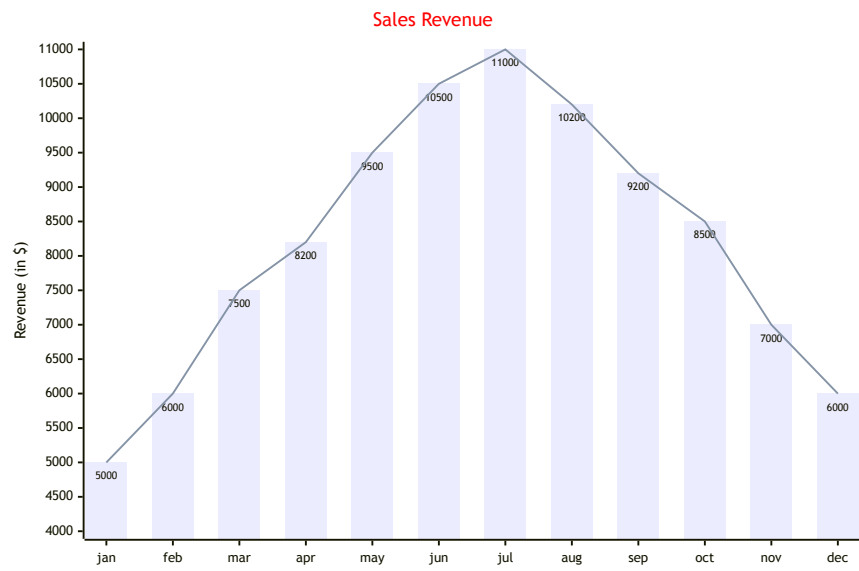
Example on config and theme

Code:

mermaid

```
---
config:
  xyChart:
    width: 900
    height: 600
    showDataLabel: true
  themeVariables:
    xyChart:
      titleColor: "#ff0000"
---
xychart
  title "Sales Revenue"
  x-axis [jan, feb, mar, apr, may, jun, jul, aug, sep, oct, nov, dec]
  y-axis "Revenue (in $)" 4000 --> 11000
  bar [5000, 6000, 7500, 8200, 9500, 10500, 11000, 10200, 9200, 8500, 7500, 6000]
  line [5000, 6000, 7500, 8200, 9500, 10500, 11000, 10200, 9200, 8500, 7500, 6000]
```

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